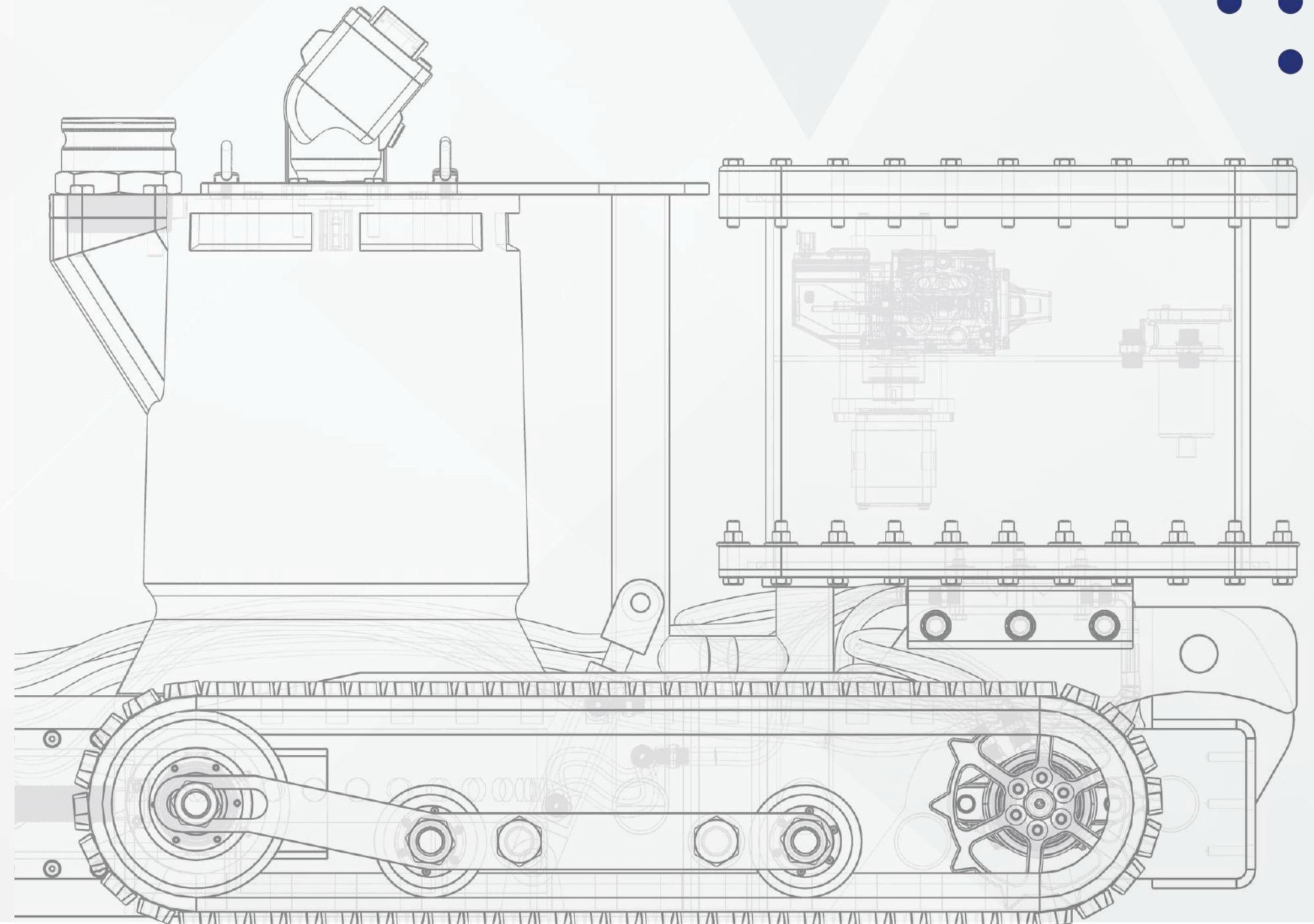




SUBMERSIBLE DREDGING VEHICLE

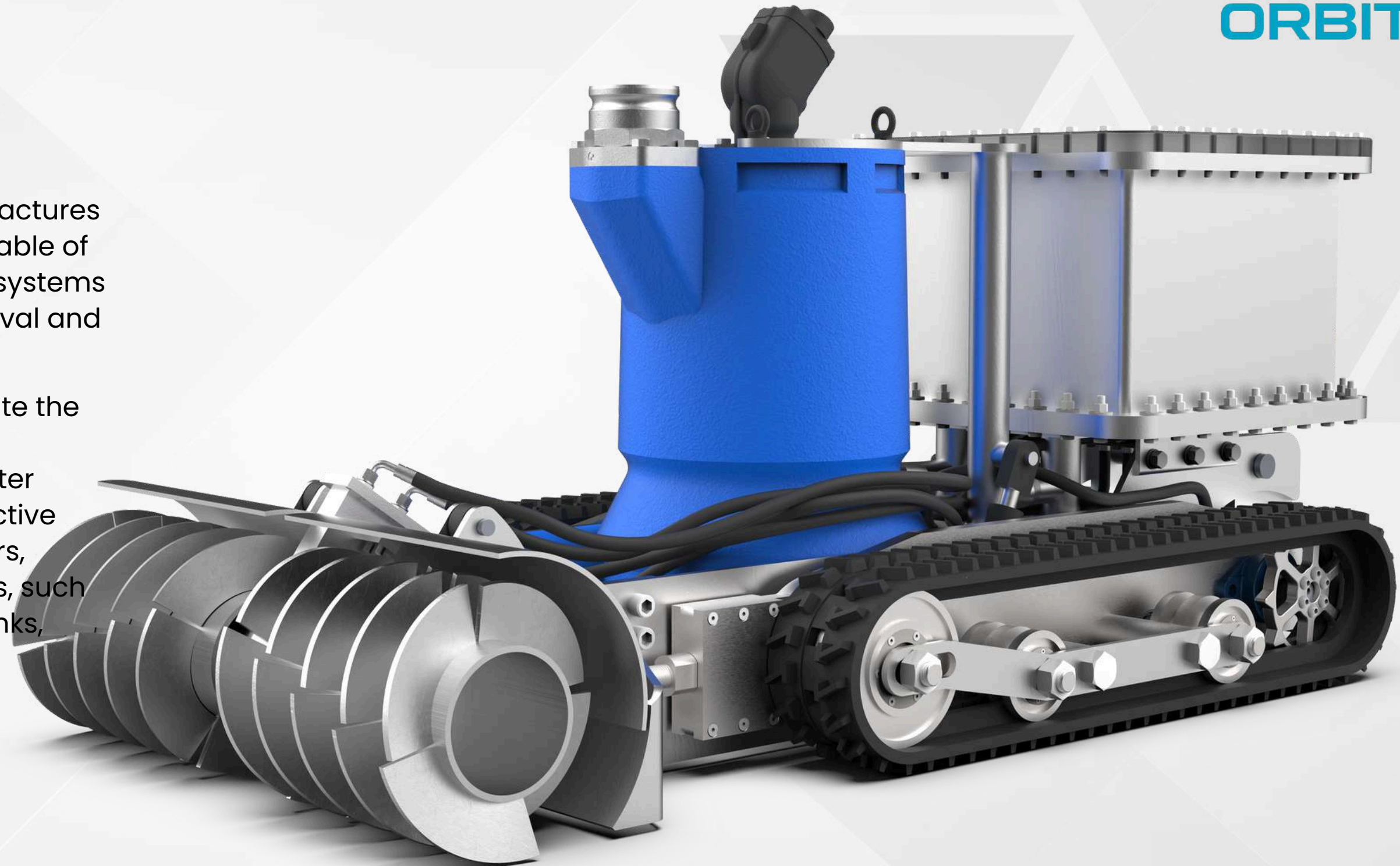
NO MAN ENTRY



INTRODUCTION

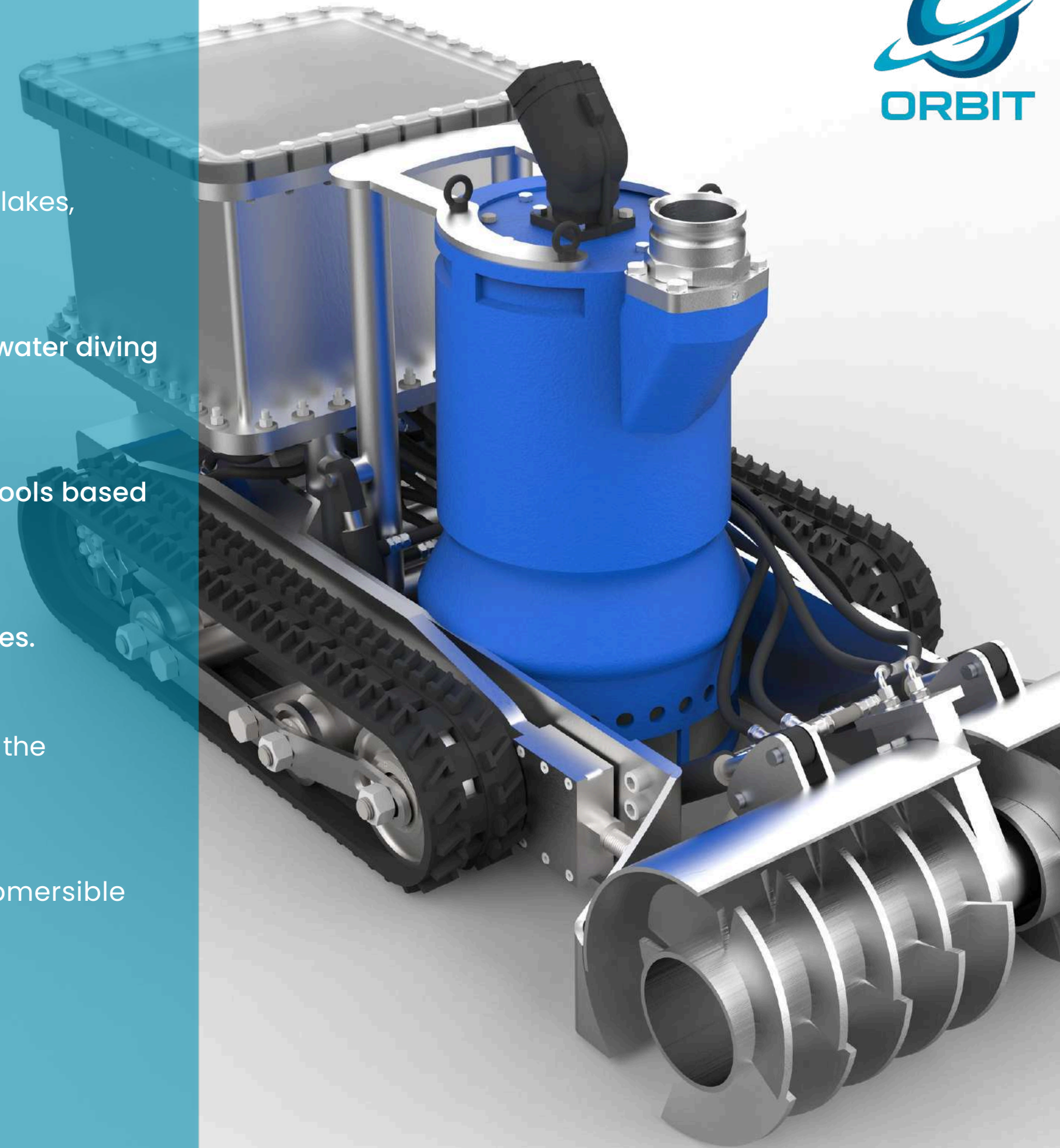


- Potential Robotic Solutions designs and manufactures advanced no-man-entry robotic systems capable of operating at depths of up to 50 meters. These systems are engineered for efficient bulk material removal and underwater cleaning applications.
- The solution is specifically developed to eliminate the need for human entry in confined and high-risk environments, including areas with extreme water pressure. It offers a safe, reliable, and cost-effective approach for removing materials from reservoirs, channels, and industrial containment structures, such as above-ground and underground storage tanks, and similar installations.



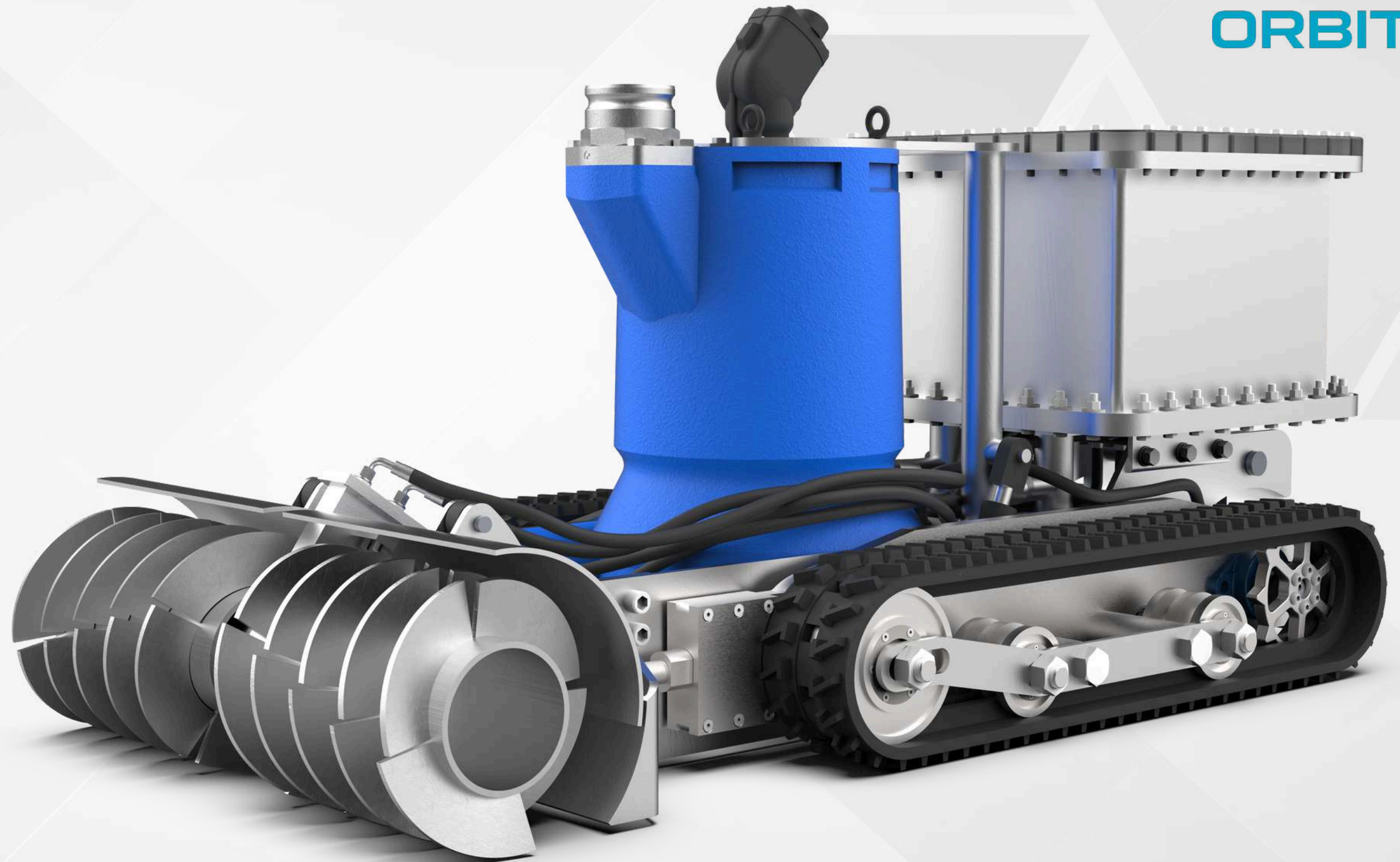
FEATURES

- **HIGH DEPTH RATED DESIGN:**
50 m Depth-Rated Design Inherently safe system suitable for reservoirs, rivers, lakes, channels, and coastal seabed operations.
- **REMOTE OPERATION:**
Implement hydraulic operated remote control to minimize human exposure to water diving for operation. This may involve using cameras for navigation and monitoring.
- **CLEANING MECHANISM:**
Adaptable cleaning mechanism using augers, rotary brushes, or specialized tools based on silt/sludge type.
- **MOBILITY:**
Designed for stable movement across uneven and confined underwater surfaces.
- **UNDERWATER VISIBILITY:**
Implement High Definition Camera and Multi-Beam Sonar visualize to results of the dredging and cleaning at very accurate visual data.
- **INBUILT SUCTION SYSTEM:**
Integrated dredging pump enabling efficient material removal within the Submersible Dredging Vehicle (SDV).

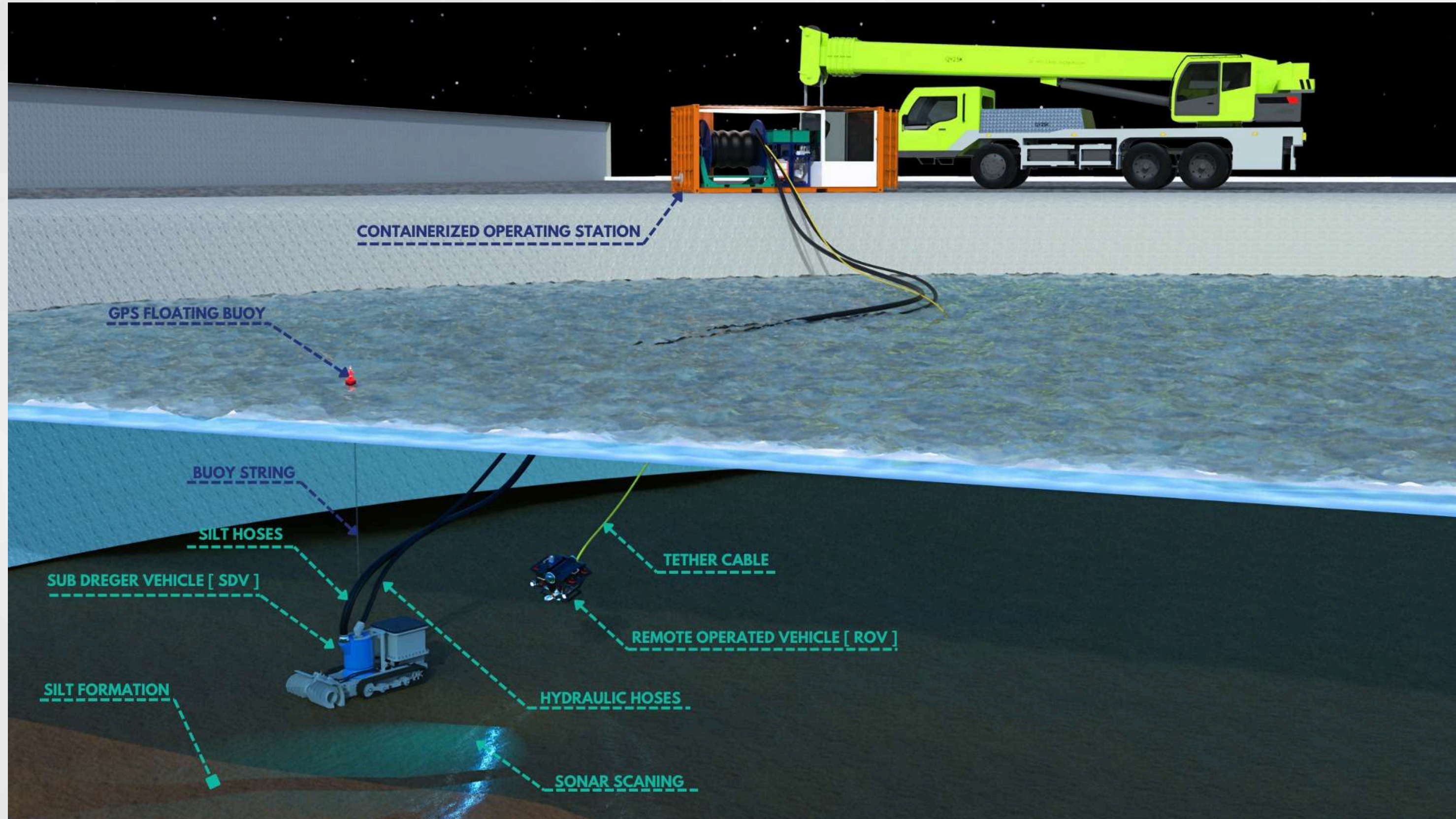


SPECIFICATIONS

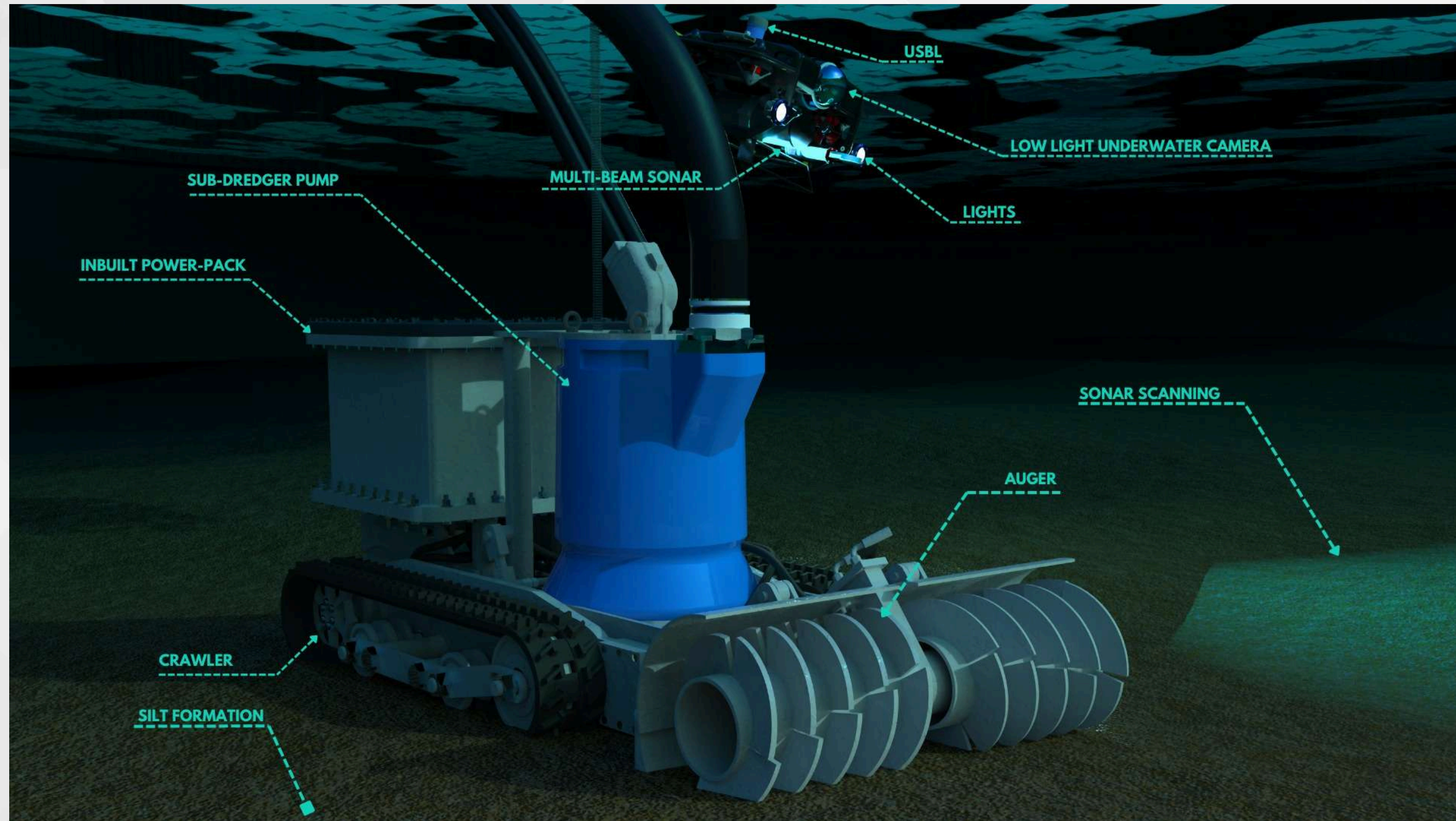
- Dimensions : 2000 x 1000 x 1000 mm
- Weight : 940 kg
- Minimum Entry Opening : 1200 X 1200 mm opening
- Operational Depth Rating : 50 m
- MOC - SS 304/ SS 316



SDV CLEANING SYSTEM



INSIDE SDV CLEANING SYSTEM



OPERATION PROCESS

1. SETUP THE OPERATION STATION

- Lower the Operational Station and Maintenance Station Containers on the site using cranes.

2. DEPLOYMENT OF THE ROBOT

- Position the dredging robot at the entry point near the reservoir.
- Lower the SDV & ROV into the water, using cranes & tethered cables as necessary.

3. NAVIGATION AND POSITIONING

- Use GPS, sonar, and cameras to guide the robot to target sediment areas.
- Continuously monitor and adjust the robot's position using remote or automated control.

4. SEDIMENT SUCTION AND REMOVAL

- Engage suction or cutter head to break up sediment and debris on the reservoir bed.
- Vacuum up loosened sediment, transferring it to an onboard or external storage system.

5. SEDIMENT TRANSPORT

- Pump dredged material through a pipeline or container to a designated disposal or treatment area.
- Regularly monitor sediment flow to prevent clogs or disruptions.

6. REAL-TIME MONITORING AND ADJUSTMENT

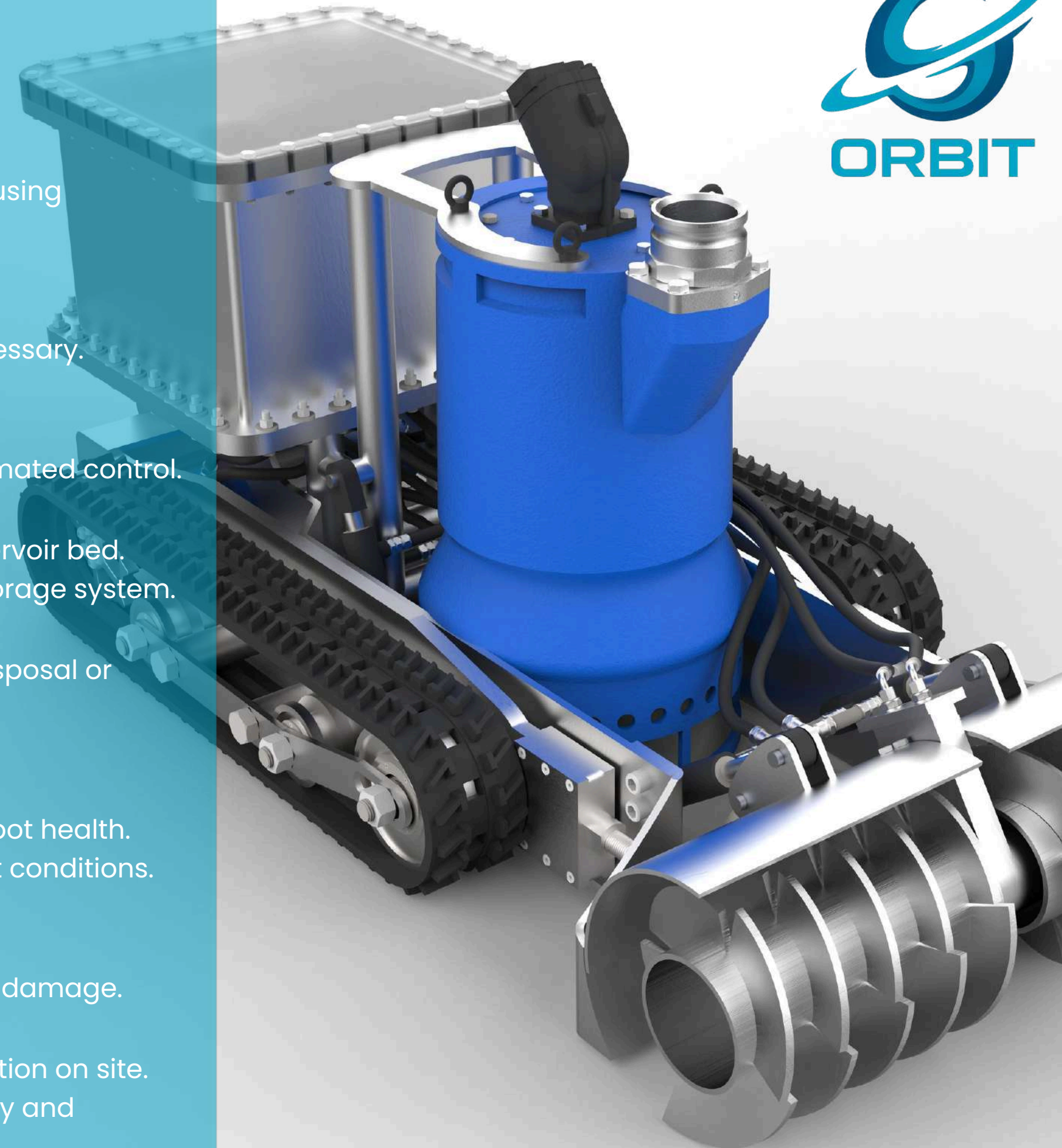
- Use sensors for real-time data on sediment density, water turbidity, and robot health.
- Make adjustments to dredging speed and suction force to match sediment conditions.

7. END-OF-OPERATION PROCEDURES

- Gradually retract the robot to the surface after completing dredging.
- Inspect and clean the robot for future deployment, addressing any wear or damage.

8. DATA COLLECTION AND REPORTING

- Using the Floating GPS Bouy, can map the operation and total work completion on site.
- Compile sediment and water data for analysis to assess dredging efficiency and environmental impact.



CONTAINERIZED OPERATING STATION



- The system includes a dedicated control room designed with ergonomic seating, allowing adjustable positioning for comfortable and efficient access to all controls.
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ACCESSORIES

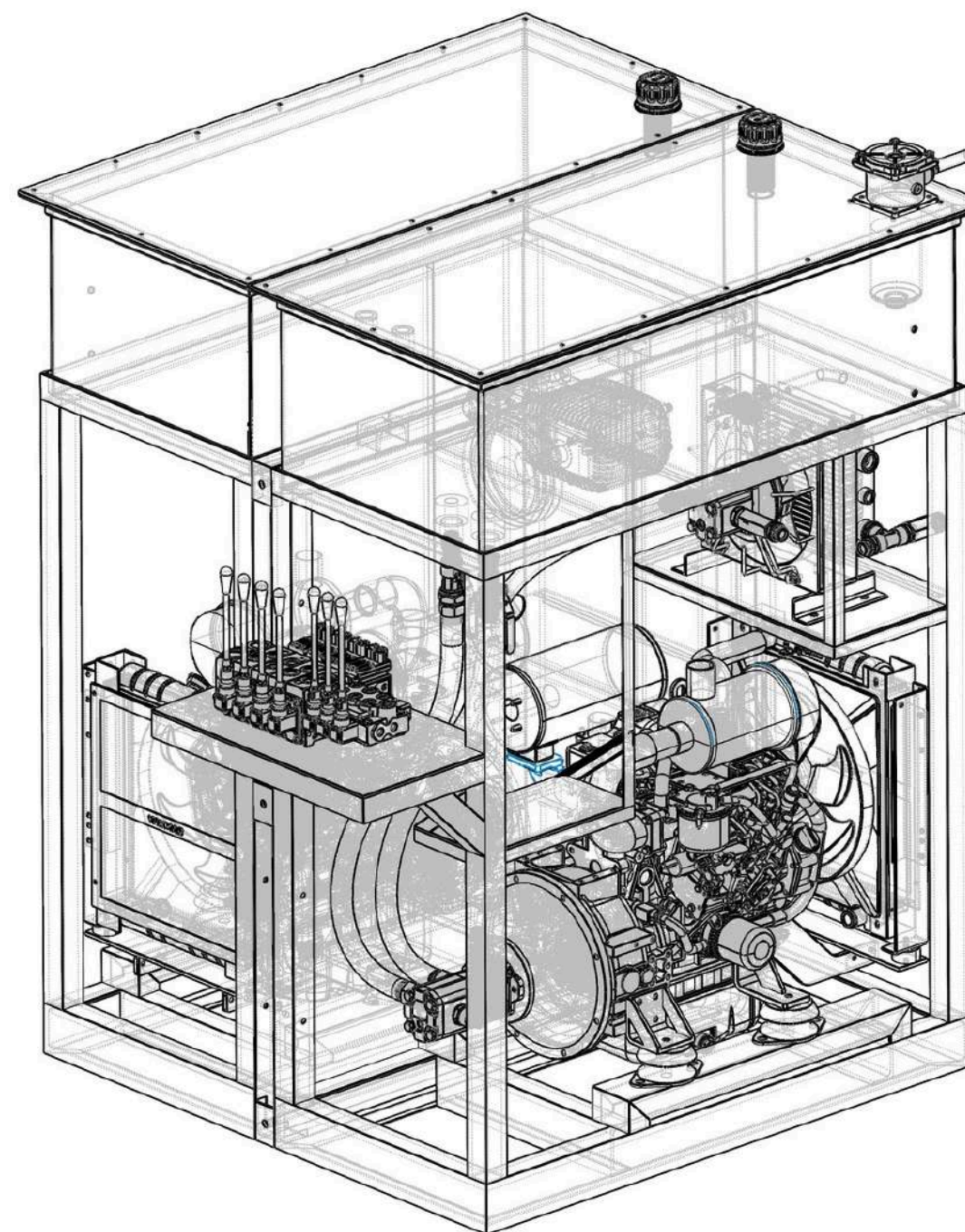
AUGER: A stainless steel, toothed auger blade designed for submersible operations. It is effective in soil, wet chemical, and acidic environments, enabling efficient breaking and crushing of sludge, silt, and muddy materials.



BRUSHWHEEL: For a well finished surface Brush wheel is used, collected silt or grit is transferred through the Dredger pump and it is transferred outside the working area. The Brush Skimmer is robust, economical, and versatile equipment suitable for a range of silt recovery applications inside reservoir, channels, Dam gates, and tanks.

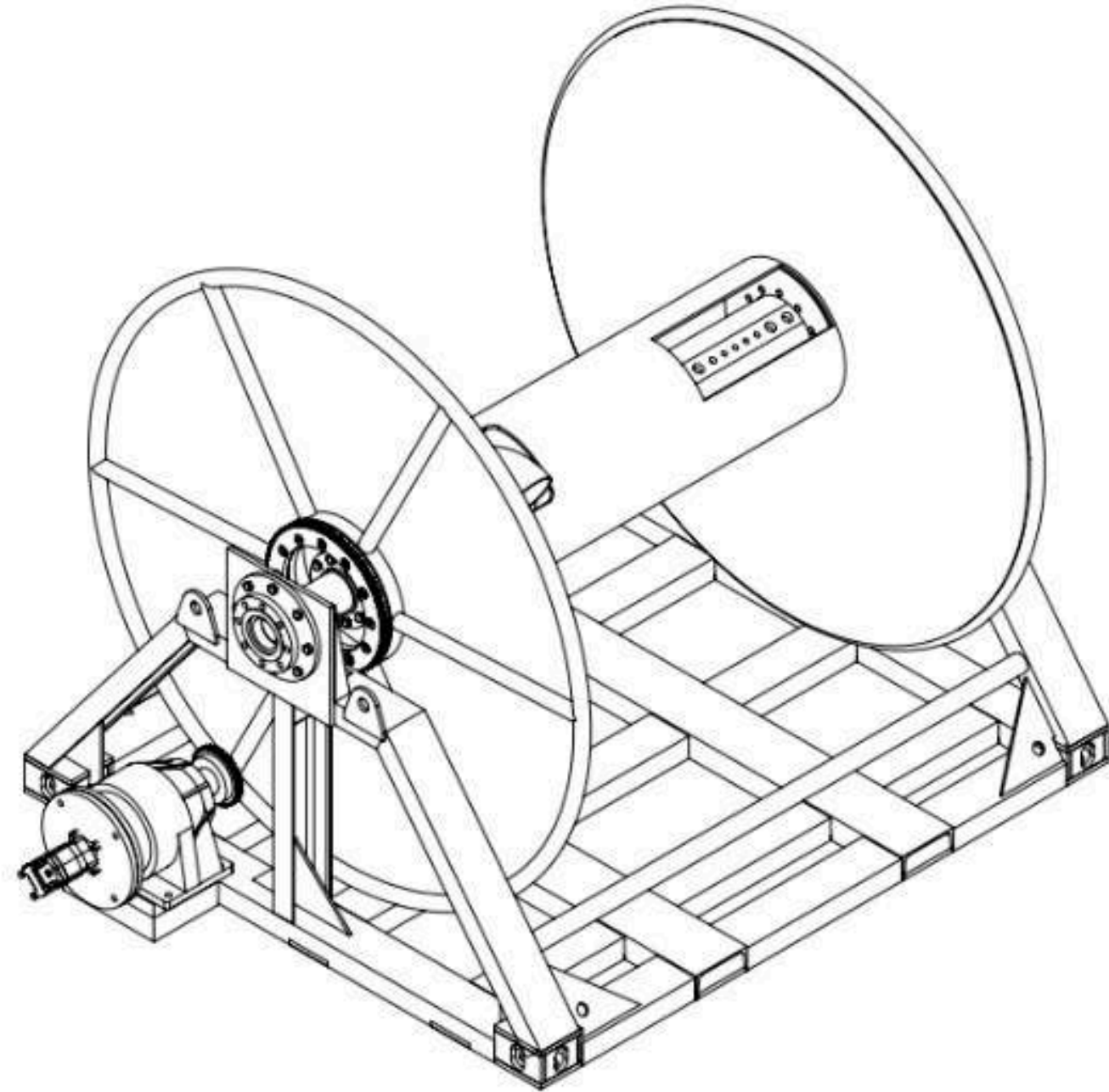
DUAL ENGINE POWER PACK

- Dual Engine Power pack is an extremely versatile unit. This power pack is offered with two or more hydraulic circuits.
- Our SDV products have been adapted to work with this power pack and therefore reduce the number of single function power packs to an absolute minimum.
- This is functioned for dredger Pump, SDV which carries Dredger Pump and Vacuum + Hydraulic Hose Reel.



SLUDGE + HYDRAULIC HOSE REEL

- The Hose Reel system consists of a robust hose reel designed for an umbilical hose, which includes hydraulic and sludge transfer hoses. It is designed for harsh, rugged water reservoirs, sewage plants, onshore and offshore areas, and is operated from a vessel or barge.



HYDRAULIC RADIO CONTROLLER

- The radio remote controller for Submersible Dredger Vehicle offers precise, remote operation of hydraulic functions, enhancing safety and efficiency.
- With this advanced controller, seamlessly manage the dredger's hydraulic systems from a distance, ensuring optimal control and maneuverability in underwater environments and challenging conditions.



GPS FLOAT BOUY

- The GPS float buoy for Submersible Dredger Vehicle is crucial for daily silt removal operations.
- It provides real-time location tracking and precise navigation, ensuring efficient and accurate dredging.
- This reliable tool enhances operational control and safety, facilitating effective silt management in various aquatic environments.



REMOTE OPERATED VEHICLE [ROV]

- The potential ROV is a remarkable underwater robot, plays a vital role in Dredging operations.
- With its advanced navigation, it deftly maneuvers through challenging waters, locating and inspecting in submerged environments.
- Additionally, its detailed inspections help assess damage to underwater structures, facilitating prompt repairs.
- This would be useful for gathering all the operational data and sonar visuals at remote locations. Its multifunctional features make it an indispensable tool for navigating underwater environments and carrying out silt removal survey.



GROUND CONTROL STATION



- The GCS is a versatile, universal, and radio-agnostic ground control platform designed for the operation of professional robotic solutions .
- It offers performance, durability, and a modular platform enabling a wide range of customizations.
- Dynamic ground control station provides greater control in scenarios where mobility is just one of many important factors.
- Easy to transport and mount, providing all the control options expected from professional ground control system.

